

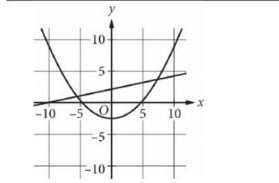
SAT Math

Nonlinear Equations and Systems 2

Question # ID
2.1 a5663025

Answer

C



A system of equations consists of a quadratic equation and a linear equation. The equations in this system are graphed in the xy -plane above. How many solutions does this system have?

- A. 0
- B. 1
- C. 2
- D. 3

2.2 d0a7871e

$$\begin{aligned} y &= x + 1 \\ y &= x^2 + x \end{aligned}$$

A

If (x, y) is a solution to the system of equations above, which of the following could be the value of x ?

- A. -1
- B. 0
- C. 2
- D. 3

2.3 7f81d0c3

$$x^2 - x - 1 = 0$$

C

What values satisfy the equation above?

- A. $x = 1$ and $x = 2$
- B. $x = -\frac{1}{2}$ and $x = \frac{3}{2}$
- C. $x = \frac{1+\sqrt{5}}{2}$ and $x = \frac{1-\sqrt{5}}{2}$
- D. $x = \frac{-1+\sqrt{5}}{2}$ and $x = \frac{-1-\sqrt{5}}{2}$

SAT Math

Nonlinear Equations and Systems 2

Question # ID

2.4 911383f2

$$(x-4)(x+2)(x-1)=0$$

Answer

D

What is the product of the solutions to the given equation?

- A. 8
- B. 3
- C. -3
- D. -8

2.5 b80d10d7

$$\frac{2(x+1)}{x+5} = 1 - \frac{1}{x+5}$$

B

What is the solution to the equation above?

- A. 0
- B. 2
- C. 3
- D. 5

2.6 fcd87b7

$$\begin{aligned}y &= x^2 - 4x + 4 \\ y &= 4 - x\end{aligned}$$

0, 3

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

SAT Math

Nonlinear Equations and Systems 2

Question #	ID		Answer
2.7	652054da	An oceanographer uses the equation $s = \frac{3}{2}p$ to model the speed s, in knots, of an ocean wave, where p represents the period of the wave, in seconds. Which of the following represents the period of the wave in terms of the speed of the wave? A. $p = \frac{2}{3}s$ B. $p = \frac{3}{2}s$ C. $p = \frac{2}{3} + s$ D. $p = \frac{3}{2} + s$	A
2.8	6e02cd78	In the xy-plane, what is the y-coordinate of the point of intersection of the graphs of $y = (x - 1)^2$ and $y = 2x - 3$?	1
2.9	802549ac	$(x + 2)(x + 3) = (x - 2)(x - 3) + 10$ Which of the following is a solution to the given equation? A. 1 B. 0 C. -2 D. -5	A
2.10	a4f61d75	$x^2 - ax + 12 = 0$ In the equation above, a is a constant and $a > 0$. If the equation has two integer solutions, what is a possible value of a?	7, 8, 13

SAT Math

Nonlinear Equations and Systems 2

Question # ID

2.11 630897df

Answer

D

The speed of sound in dry air, v , can be modeled by the formula $v = 331.3 + 0.606T$, where T is the temperature in degrees Celsius and v is measured in meters per second. Which of the following correctly expresses T in terms of v ?

A. $T = \frac{v + 0.606}{331.3}$

B. $T = \frac{v - 0.606}{331.3}$

C. $T = \frac{v + 331.3}{0.606}$

D. $T = \frac{v - 331.3}{0.606}$

2.12 c77ef2fb

A

Blood volume, V_B , in a human can be determined using the equation

$$V_B = \frac{V_P}{1 - H},$$

where V_P is the plasma volume and H is the hematocrit (the

fraction of blood volume that is red blood cells). Which of the following correctly expresses the hematocrit in terms of the blood volume and the plasma volume?

A. $H = 1 - \frac{V_P}{V_B}$

B. $H = \frac{V_B}{V_P}$

C. $H = 1 + \frac{V_B}{V_P}$

D. $H = V_B - V_P$

2.13 364a2d25

8, 9

$$\begin{aligned} x + y &= 17 \\ xy &= 72 \end{aligned}$$

If one solution to the system of equations above is (x, y) , what is one possible value of x ?

SAT Math

Nonlinear Equations and Systems 2

Question # ID
2.14 0980fcdd

$$\begin{aligned}x^2 &= 6x + y \\ y &= -6x + 36\end{aligned}$$

A solution to the given system of equations is (x, y) . Which of the following is a possible value of xy ?

- A. 0
- B. 6
- C. 12
- D. 36

Answer

A

2.15 87a3de81

$$x^2 + x - 12 = 0$$

If a is a solution of the equation above and $a > 0$, what is the value of a ?

3

2.16 2683b5db

$$T = 0.01(P - 40,000)$$

In a city, the property tax T , in dollars, is calculated using the formula above, where P is the value of the property, in dollars. Which of the following expresses the value of the property in terms of the property tax?

- A. $P = 100T - 400$
- B. $P = 100T + 400$
- C. $P = 100T - 40,000$
- D. $P = 100T + 40,000$

D

SAT Math

Nonlinear Equations and Systems 2

Question # ID
2.17 2f958af9

Answer

A

$$v^2 = \frac{LT}{m}$$

The formula above expresses the square of the speed v of a wave moving along a string in terms of tension T , mass m , and length L of the string. What is T in terms of m , v , and L ?

A. $T = \frac{mv^2}{L}$

B. $T = \frac{m}{v^2L}$

C. $T = \frac{mL}{v^2}$

D. $T = \frac{L}{mv^2}$

2.18 876a731c

$y = x^2$
$2y + 6 = 2(x + 3)$

A

If (x, y) is a solution of the system of equations above and $x > 0$, what is the value of xy ?

A. 1

B. 2

C. 3

D. 9

2.19 928498f3

$$6x^2 + 5x - 7 = 0$$

A

What are the solutions to the given equation?

A. $\frac{-5 \pm \sqrt{25 + 168}}{12}$

B. $\frac{-6 \pm \sqrt{25 + 168}}{12}$

C. $\frac{-5 \pm \sqrt{36 - 168}}{12}$

D. $\frac{-6 \pm \sqrt{36 - 168}}{12}$

SAT Math

Nonlinear Equations and Systems 2

Question # ID

2.20 2d2ab76b

$$y = x^2 - 1$$

$$y = 3$$

When the equations above are graphed in the xy -plane, what are the coordinates (x, y) of the points of intersection of the two graphs?

- A. $(2,3)$
and $(-2,3)$
- B. $(2,4)$
and $(-2,4)$
- C. $(3,8)$
and $(-3,8)$
- D. $(\sqrt{2},3)$
and $(-\sqrt{2},3)$

Answer

A

2.21 3b4b8831

$$38x^2 = 38(9)$$

What is the negative solution to the given equation?

-3

2.22 f5247e52

$$y = ax^2 - c$$

In the equation above, a and c are positive constants. How many times does the graph of the equation above intersect the graph of the equation $y = a + c$ in the xy -plane?

- A. Zero
- B. One
- C. Two
- D. More than two

C

2.23 f76c1858

$$7x^2 - 20x - 32 = 0$$

What is the positive solution to the given equation?

4

SAT Math

Nonlinear Equations and Systems 2

Question # ID

Answer

2.24 4e18fc5d

A

$$v = -\frac{w}{150x}$$

The given equation relates the distinct positive numbers v , w , and x . Which equation correctly expresses w in terms of v and x ?

A. $w = -150vx$

B. $w = -\frac{150v}{x}$

C. $w = -\frac{x}{150v}$

D. $w = v + 150x$

2.25 bef4b1c6

5

$$\frac{55}{x+6} = x$$

What is the positive solution to the given equation?

2.26 11ccf3e1

A

$$14j + 5k = m$$

The given equation relates the numbers j , k , and m . Which equation correctly expresses k in terms of j and m ?

A. $k = \frac{m-14j}{5}$

B. $k = \frac{1}{5}m - 14j$

C. $k = \frac{14j-m}{5}$

D. $k = 5m - 14j$

2.27 29ed5d39

D

$$p = 9 + \frac{14}{n}$$

The given equation relates the numbers p and n , where n is not equal to 0 and $p > 9$. Which equation correctly expresses n in terms of p ?

A. $n = \frac{p-9}{14}$

B. $n = \frac{p}{14} + 9$

C. $n = \frac{p}{14} - 9$

D. $n = \frac{14}{p-9}$

SAT Math

Nonlinear Equations and Systems 2

Question # ID
2.28 895628b5

Answer

$$\begin{aligned}y &= (x - 2)(x + 4) \\y &= 6x - 12\end{aligned}$$

C

Which ordered pair (x, y) is the solution to the given system of equations?

- A. $(0, 2)$
- B. $(-4, 2)$
- C. $(2, 0)$
- D. $(2, -4)$

2.29 13e57f0a

B

$$-4x^2 - 7x = -36$$

What is the positive solution to the given equation?

- A. $\frac{7}{4}$
- B. $\frac{9}{4}$
- C. 4
- D. 7

2.30 6acdcece

D

$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

- A. $x = \frac{b-72}{y}$
- B. $x = by - 72$
- C. $x = \frac{by-72}{y}$
- D. $x = by - 72y$

2.31 09e5e4d3

C

If $\frac{42}{x} = 7x$, what is the value of $7x^2$?

- A. 6
- B. 7
- C. 42
- D. 294

SAT Math

Nonlinear Equations and Systems 2

Question # ID
2.32 4b6f0a3f

Answer

5

$$x^2 - 5x - 24 = 0$$

What is the sum of the solutions to the given equation?

2.33 75a32330

A

$$y = x^2 + 1.7$$

$$y = 1.7 - x$$

Which graph represents the given system of equations?

